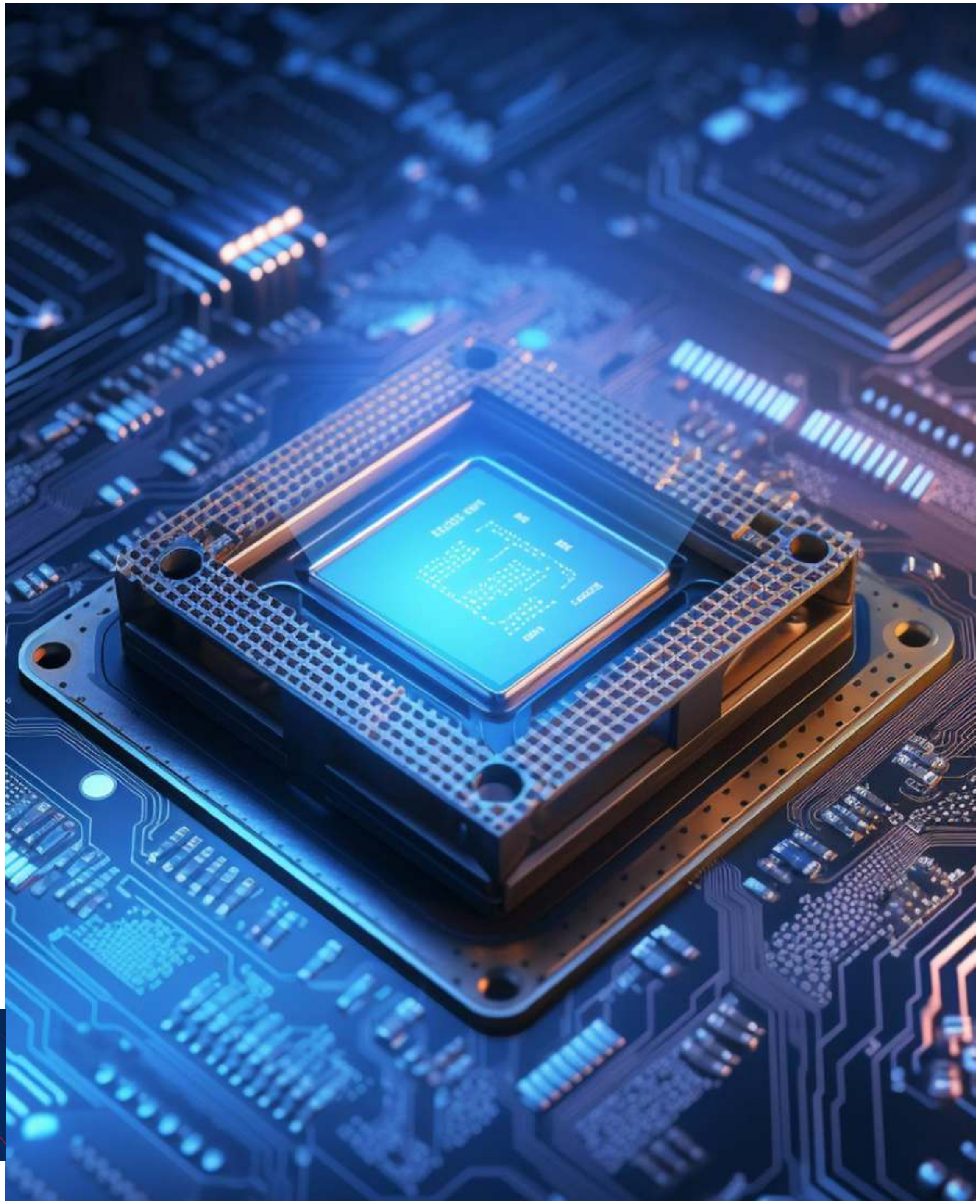
इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY

सत्यमेव जयते

इंडिया सेमीकंडक्टर मिशन
India Semiconductor Mission
Catalyzing India's Semiconductor Ecosystem

Semiconductor Manufacturing and Display Ecosystem In India

Semicon India Program



Vision for a New India

Leading global value chains through Hi-tech manufacturing

“

India is committed to becoming *the world’s reliable partner in global supply chains.* This is the best time to invest in India.

”

“

Our dream is that every device in the world will have an Indian-made chip

Semicon India 2024

”

“

India is making policies keeping in mind the *goals of the next 25 years.*

‘State of World’ address World Economic Forum, 2022

”

Shri Narendra Modi
Hon’ble Prime Minister of India

Digital Public Infrastructure – Driving New Markets

Fastest Growing
G-20 Economy

UPI UNIFIED PAYMENTS INTERFACE

Global Leader
in the Fast Payment

Digital India
Power To Empower

969+ Mn
Internet Users
till March '25

#startupindia

3rd largest
Startup Ecosystem

1.16 Billion
Mobile Subscribers
till March '25

Highest
Fin-Tech adoption

~US$30 Bn in Fiscal Support

Support to Make India Global Hub for Electronics Manufacturing

Incentive Outlay ~$10 Bn

Support for Semiconductor and Display Ecosystem

1. Semiconductor Fabs and Display Fabs
2. Compound Semiconductor and ATMP
3. Design Linked Incentive (DLI)
4. Modernization of Semi-conductor Laboratory (SCL)

Incentive Outlay ~$10 Bn

Support for Electronics Manufacturing

1. PLI for Mobiles & Components,
2. PLI for IT Hardware
3. Electronic Component Manufacturing Scheme
4. Development of Electronics Manufacturing Clusters
5. Manufacturing of Electronic Components & Semiconductor

Incentive Outlay ~$10 Bn

Support for Allied Sectors

Production Linked Incentives for

1. Automobiles & Auto Components
2. Advanced Chemistry Cell
3. Solar PV Modules
4. Telecom & Networking
5. White Goods
6. Medical Devices
7. Drones & Components

India's Electronics Manufacturing : Overview (Last Decade)

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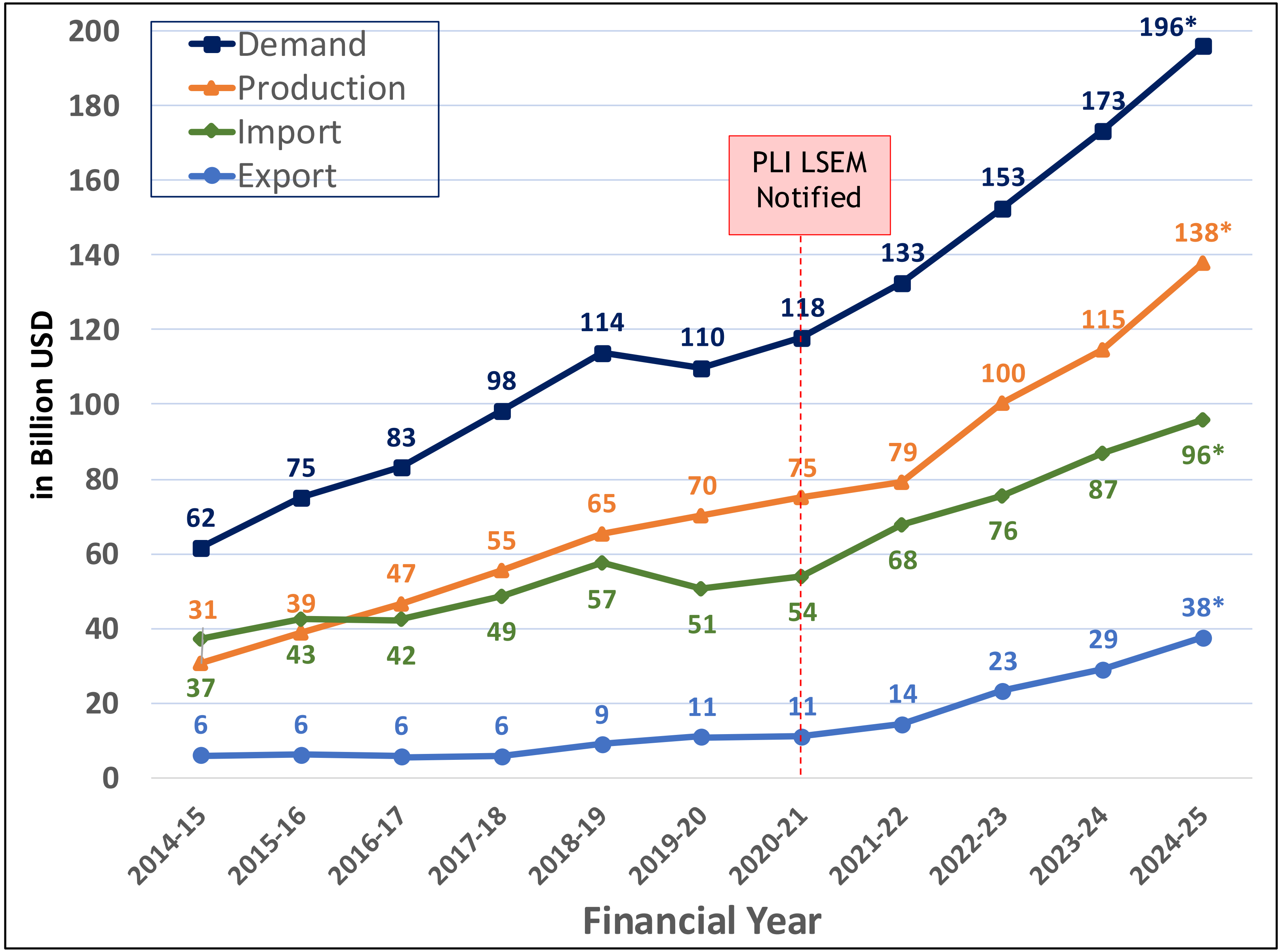


Source: Industry and DGFT (*Data for FY2024-25 - Industry Estimates)

Electronics Manufacturing (FY 2024-25)

- **Production:** Expected ↑ **6X**
- **Exports:** Expected ↑ **8X**
- Production Mobile ↑ **28X**
- Production Non-mobile (Exp) **↑130X**

Government Support for Development of Semiconductors & Display Ecosystem

Semiconductor & Display Fabs

- **Fiscal Support:** 50% of eligible project cost on pari-passu basis
- **Additional Support:** Infrastructure Support, Demand Aggregation, R&D & Skill Development

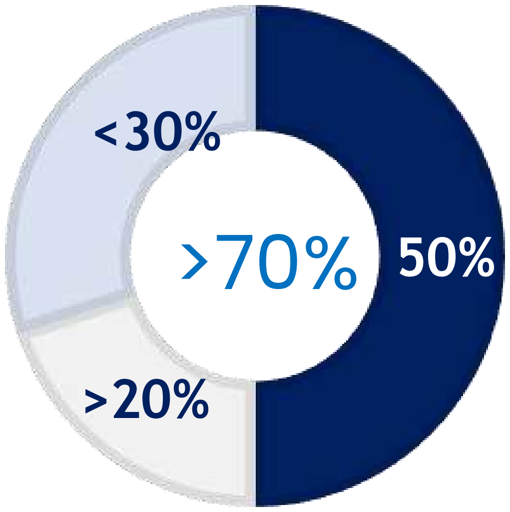
Compound Semiconductor & Packaging

- **Fiscal Support:** 50% of Capex on pari-passu basis
- **Additional Support:** Infrastructure Support, Demand Aggregation, R&D & Skill Development

Semiconductor Design

- **Product Design Linked Incentive:** up to 50% of eligible Capex (ceiling INR 15 Cr/ USD 1.8 Mn)
- **Deployment Linked Incentive:** 4% - 6% on Net Sales

50% pari-passu Incentives for Semicon Manufacturing (~$10 Bn Outlay)



Government of India (Pari-Passu)

State Government

Applicant

Semiconductor Nodal Agency: India Semiconductor Mission

Fabs	Display Fabs	Compound Semiconductors	ATMP / OSAT	Design

Approvals under Semicon India Programme (10)

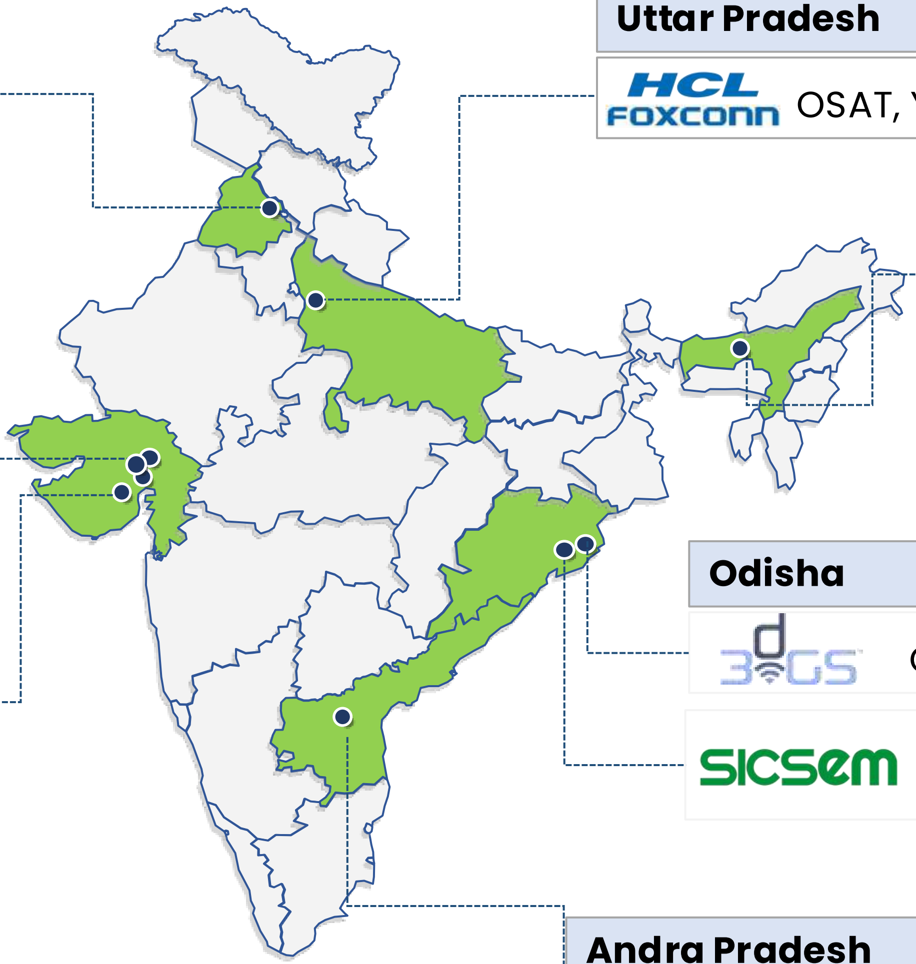
*Semiconductor **Fab***	*Outsourced Semiconductor Assembly & Test (OSAT)/ATMP*			
TATA / PSMC	Micron	TATA	CG / RENESAS / STARS MICROELECTRONICS	KAYNES TECHNOLOGY
Dholera, GJ	**Sanand, Gujarat**	**Jagiroad, Assam**	**Sanand, Gujarat**	**Sanand, Gujarat**
~USD 11 Bn Investment v	**~USD 2.7 Bn** Investment	**~USD 3.25 Bn** Investment	**~USD 915 Mn** Investment	**~USD 400 Mn** Investment
50,000 WSPM Capacity	**~3.8 Mn units/day** Capacity	**~48 Mn units/day** Capacity	**~15 Mn units/day** Capacity	**~6.33 Mn units/day** Capacity
28/45/55/90/110 nm nodes				

Approvals under Semicon India Programme

Compound Fab, Outsourced Semiconductor Assembly & Test (OSAT)/ATMP

SiCSem	HCL Foxconn	ASIP Technologies	CDIL Semiconductors	3D GS
Bhubaneswar **Odisha**	**YEIDA, Uttar Pradesh**	**Andhra Pradesh**	**Mohali, Punjab**	Bhubaneswar **Odisha**
~USD 250 Mn Investment	**~USD 445 Mn** Investment	**~USD 57 Mn** Investment	**~USD 14 Mn** Investment	**~USD 235 Mn** Investment
5,000 WSPM Capacity	**~1.2 Mn units/day** Capacity	**~96 Mn/Yr** Capacity	**~158 Mn /Yr** Capacity	**~50 Mn /Yr** Capacity

Ten Approved Projects



Punjab

CDIL OSAT, Mohali

Uttar Pradesh

HCL FOXCONN OSAT, YEIDA

Assam

TATA TATA OSAT, Jagiroad

Gujarat

micron ATMP, Sanand

G OSAT, Sanand

KAYNES TECHNOLOGY OSAT, Sanand

TATA Tata Fab, Dholera SIR

Odisha

3dGS OSAT, Bhubaneswar

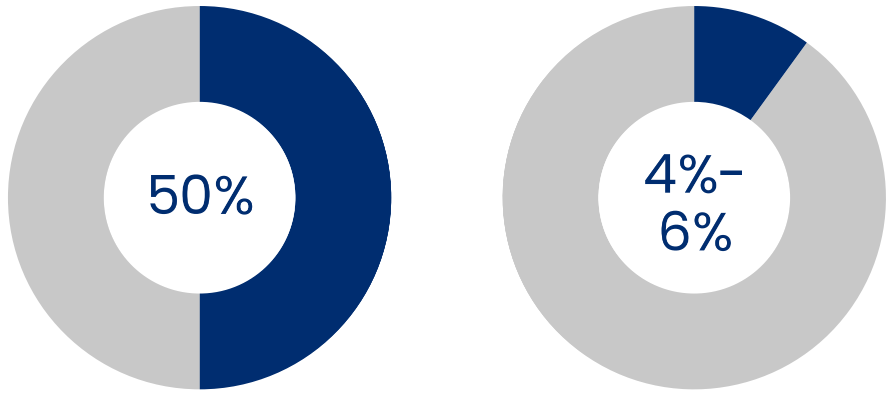
SiCSem Compound Fab & ATMP, Bhubaneswar

Andra Pradesh

ASIP TECHNOLOGIES OSAT

Design-linked Incentive (DLI) scheme

Product Design Linked | **Deployment Linked**



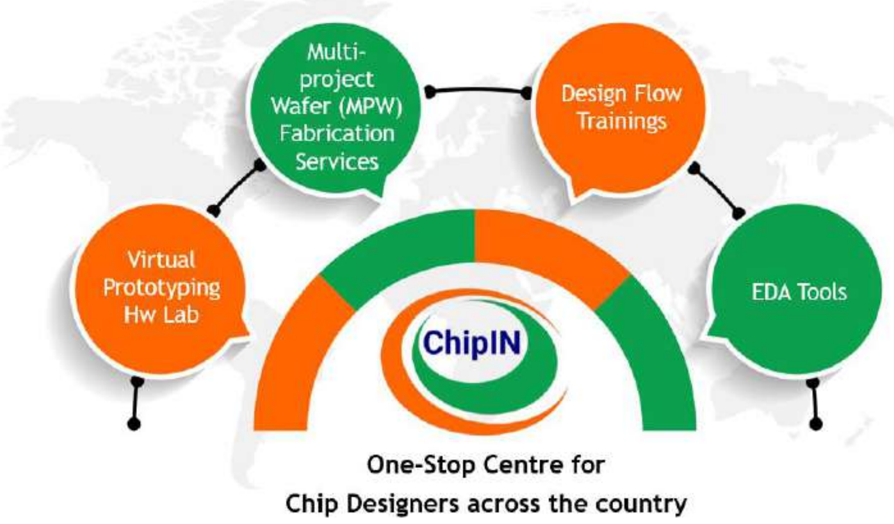
Government of India   Applicant

Infrastructure Support – Free EDA Tools

24 DLI Startup Design Companies approved

87 Startups approved for EDA Tools Support

13.5 Million Hours usage of EDA Tools in 2025



Under **Chips to Startups (C2S) Programme**: EDA Tools to **285** Academic Institutions for training **85K** specialized manpower

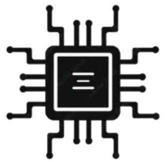
Important Chips/ System on Chips

CCTV Chips

Netrasemi

3rdiTech

MINDGROVE



Microprocessors

InCore

MORPHING MACHINES

Communications Infra

Aheesa DIGITAL INNOVATIONS

WiSig Networks

MMRFIC

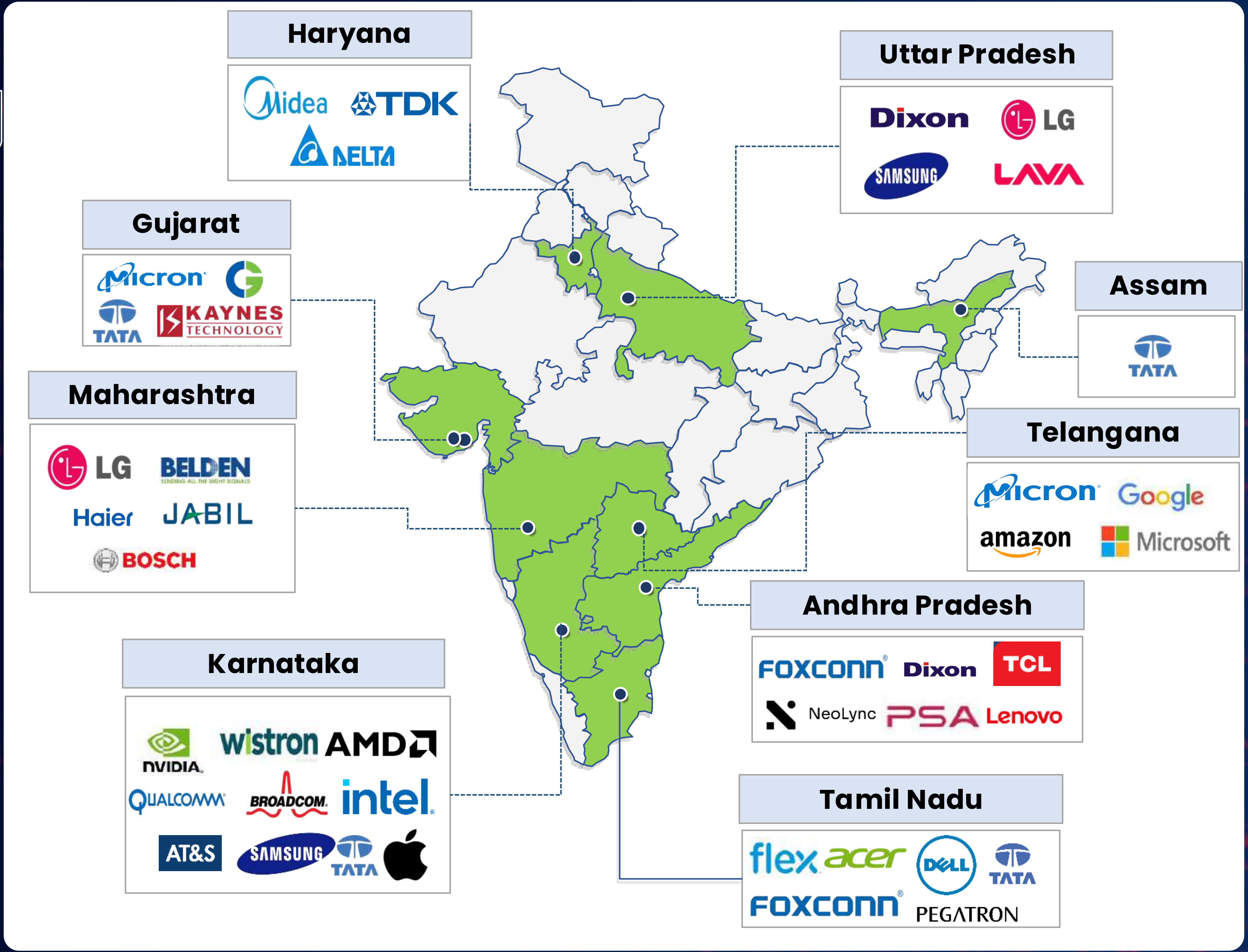
Skilling and Capacity Building Initiatives

Chips to Startup (C2S)	• Empowering the semiconductor ecosystem by training **85,000 specialized professionals** through hands-on projects and industry-linked programs.
New Semiconductor Curriculum	• A steady talent pipeline with **8,000 graduates annually** across **500+ colleges**, offering **UG, PG, and diploma** courses for semiconductor design and manufacturing.
LAM Semiverse Programme	• In collaboration with **IISc, MeitY, and ISM**, training **60,000 students over 10 years** through immersive virtual labs and industry-driven learning modules.
Applied Materials	• Investing **$400 million over four years** to establish a **R&D and engineering center in India**, catalyzing **$2 billion in investments** and generating **500 advanced engineering** and **2,500 indirect jobs** within five years.
IndiaAI FutureSkills	• Building the next generation of AI talent by supporting **8,000 undergraduates, 5,000 postgraduates, and 500 PhD fellows.**

Growing Electronics Ecosystem

Key Clusters / Hubs

1. **Uttar Pradesh –** Noida, Greater Noida
2. **Haryana –** Gurgaon, Manesar
3. **Tamil Nadu -** Sriperumbudur
4. **Andhra Pradesh –** Sri City, Tirupati
5. **Maharashtra –** Pune
6. **Karnataka -** Bangalore, Mysore
7. **Telangana -** Hyderabad
8. **Gujarat** – Dholera / Sanand
9. **Assam** - Bongora



Existing Ecosystem to Catalyze Growth

Pre-Competitive Research

CeNSE

DRDO SSPL

Centre for Materials for Electronics Technology (CMET)

Manufacturing

SCL

SITAR

GAETEC

Packaging & Services

SCL

SPEL

TESSOLVE

MEMS & Discrete

CDIL

भारत इलेक्ट्रॉनिक्स BHARAT ELECTRONICS

SITAR

Equipment / Tools

APPLIED MATERIALS

Lam RESEARCH

KLA

ADVANTEST

EDA Tools

synopsys

arm

Mentor

cadence

Design

intel

Micron

Qualcomm

MARVELL

MEDIATEK

AMD

ST life.augmented

ANALOG DEVICES

BROADCOM

NVIDIA

NXP

TEXAS INSTRUMENTS

Indian Fabless

Saankhya Labs

SIGNALCHIP

STERADIAN

CIREL SYSTEMS

Bulk Gases Companies in India & Ecosystem Development

Existing production capabilities of major fab consumables

Materials	Global benchmark for the semiconductor industry	Availability
Oxygen	Ultra-high purity – at least 99.999%	✓
Nitrogen	At least 99.99 to 99.999	✓
Hydrogen	99.99999+% purity	✓
Argon	99.999% Purity	✓
Carbon Dioxide	99.995% purity	✓

Source: IESA Report

Leading global gas suppliers have manufacturing presence in India

Potential Suppliers in India

INOX AIR PRODUCTS

Linde

Air Liquide

PRAXAIR

SUPERIOR

CARBONIC INDUSTRIES
MANUFACTURERS OF CO_2 GAS

AGEM®

INDIA GLYCOLS LIMITED

TATA STEEL

Capability to upgrade existing gas-producing facilities to semiconductor grade

Source: IESA Report

Chemical Companies In India & Ecosystem Development

Existing production capabilities of major fab consumables

Materials	Global benchmark for the semiconductor industry	Availability
Sulphuric Acid	98% Concentration	✓
Hydrochloric Acid	37-38% Concentration	✓
Ammonia	Purity level of 99.99%	✓
Hydrogen Peroxide	30% concentration, ultrapure	✓
Nitric Acid	70% Concentration	✓
Acetic Acid	99% in VLSI-quality	✓
Ethanol	70-90% concentration	✓
Acetone	>=99.5%, ULSI Grade	✓
Carbon	99.995% purity	✓
Phosphoric Acid	Greater than 99% purity	✓

3 Largest producer of Sulphuric Acid globally

4 Largest producer of Ammonia globally

Potential Suppliers in India

- HINDALCO
- Hindustan Zinc Limited
- NPL
- SATYENDRA FINECHEM
- bajaj SUGAR
- TATA CHEMICALS
- FINAR Finar Limited
- Omniscient Associates
- IFFCO
- GACL Promoting Green Technology

Source: IESA Report

Material Companies in India & Ecosystem Development

Existing production capabilities of major fab consumables

Materials	Global benchmark for the semiconductor industry	Availability
Bauxite	Al, AlSi, AlCu Composition range of selected elements	✓
Silicon/Quartz	99.9999999% Purity level	✓
Gallium	Purity greater than 99.9999%	✓
Gypsum	Above 95% purity	✓
Antimony	99.999+ percent pure	✓
Magnesium	99.9%	✓
Titanium	High-purity titanium of around 99% by mass.	✓
Zinc	99.99% pure zinc	✓
Molybdenum	99.995%	✓
Rare Earth	99.999%	✓

Potential Suppliers in India

NALCO

APMDC

UCIL

HINDALCO

Hindustan Zinc Limited

TANMAG

IREL (India) Limited

Source: IESA Report

Building Global Semiconductor Partnerships

JAPAN

- ◊ MoC on Semiconductor Supply Chain Partnership

USA

- ◊ MoU on Building Supply Chain Resilience & Innovation
- ◊ MoU with IBM & MoU with Purdue University

EU

- ◊ MoU on Semiconductor Supply Chain Partnership under the framework of Trade & Technology Council (TTC), R&D, and skills

Singapore

- ◊ MoU with Singapore on Semiconductor Supply Chain Partnership, skill and customs best practices.

Recent Announcements

APPLIED MATERIALS	$400 Mn. for setting up collaborative engineering centre
AMD	$400 Mn. in next five years to expand R&D operations in India
MICROCHIP	$300 Mn. in expanding its R&D presence in India
Lam RESEARCH	USD 1.2 billion, Sourcing components, R&D. Train 60K using Semiverse
MERCK & Fujifilm	MERCK MoU with Tata Exploring Set up VLSI Grade Chemical Manf. Units in India
TEL	Exploring Set up Equipment Training and Service Support Centre in India

Semiconductor Policies by State Governments *(Over and Above Central Government Support)*

Uttar Pradesh

25% on CAPEX

Gujarat

20% on CAPEX

Madhya Pradesh

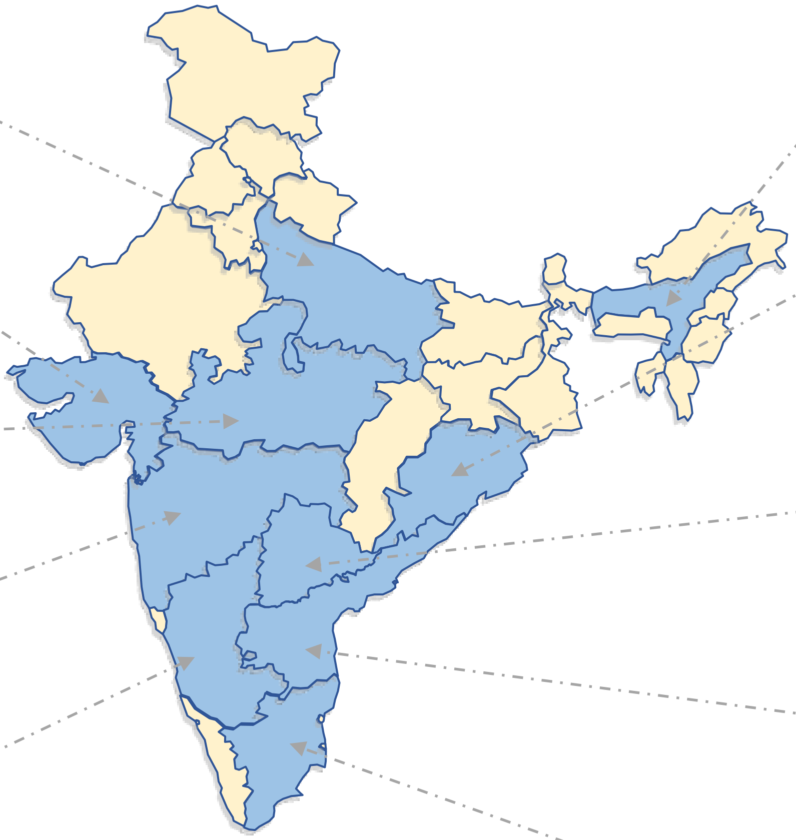
25% of CAPEX

Maharashtra

Special Incentive Package

Karnataka

25% Capital Subsidy



Assam

20% on CAPEX

Odisha

- 25% of CAPEX support
- 30-50% for non-ISM projects

Telangana

25% of CAPEX

Andhra Pradesh

30% of CAPEX

Tamil Nadu

25% of CAPEX

All states have linked their support with the Government of India- 50% pari-passu support; Additional subsidy on Land, Water, Power, etc., employee-linked incentives, Interest Subsidy, Tax Waivers, Stamp Duty Waiver, etc.

Way Forward: India Semiconductor Mission 2.0

Long term government support (10-20 yrs)

Support For
Packaging (ATMP/OSAT)

Support For
Semiconductor Fabs

Support for
R&D
Fab(Adv Nodes), Adv. Packaging

Support for
Display fabs: OLED, Micro LED, LCD

Support for Capital Equipment, Components/Sub-Assemblies

Support for
Compound Semiconductor fabs

Support for
Semiconductor grade
Gasses, Chemicals, Materials

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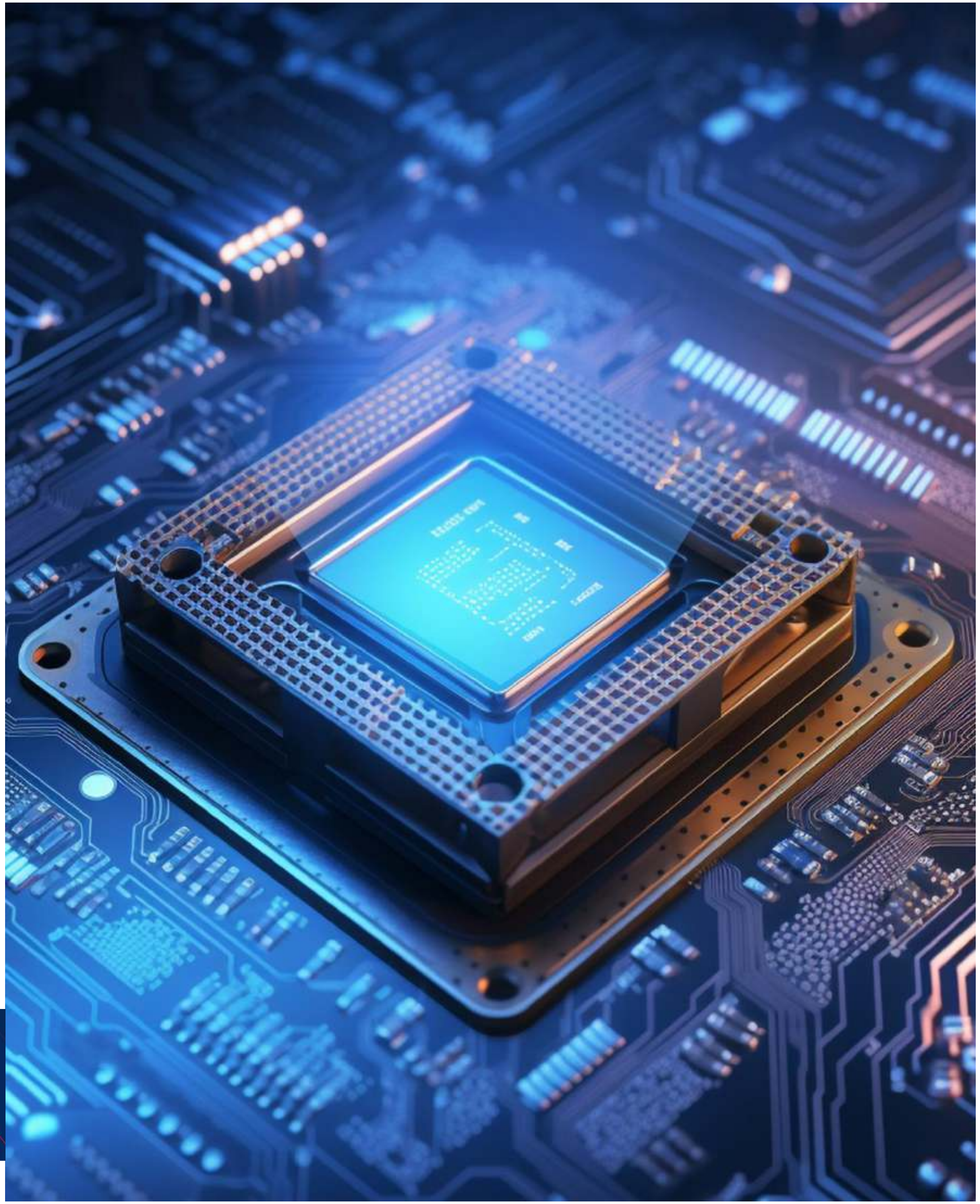
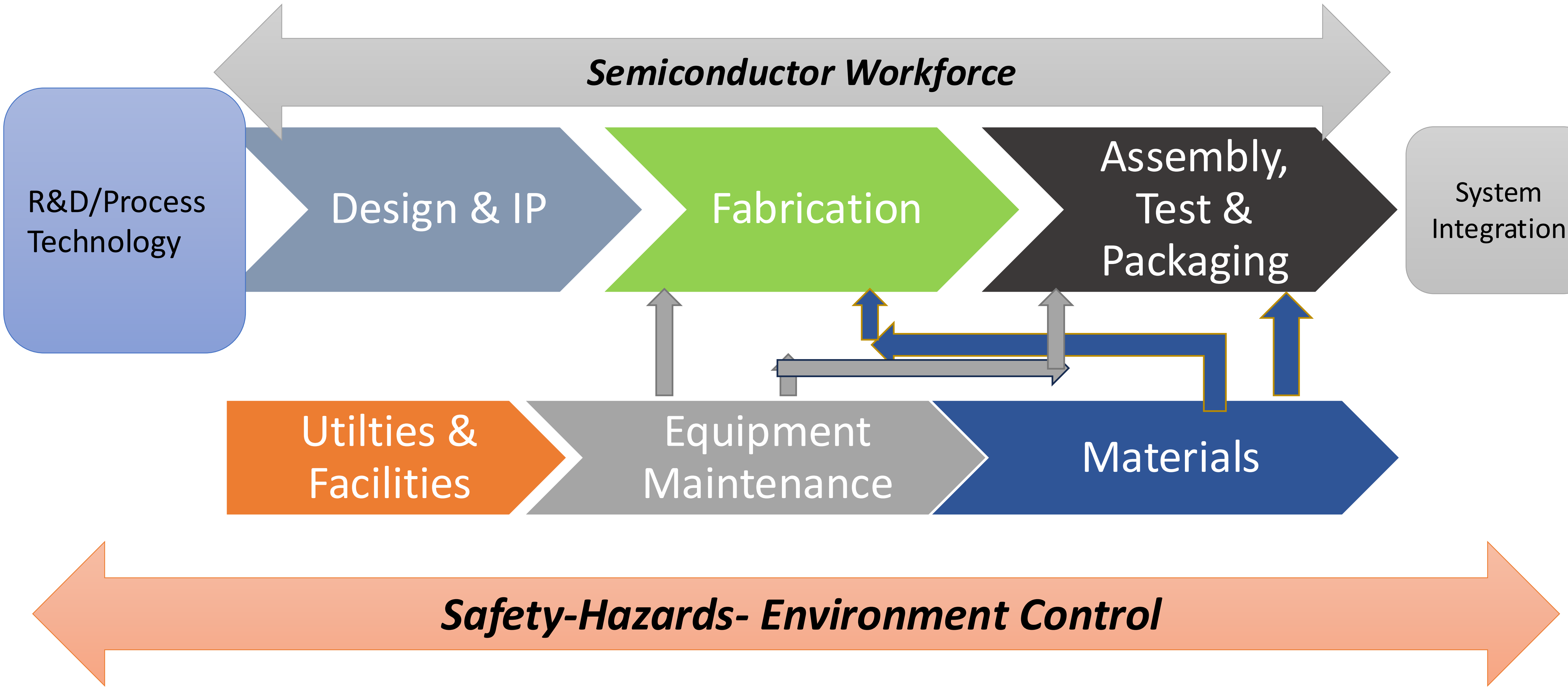
Building a Sustainable Semiconductor Workforce:

Roadmap & Approach

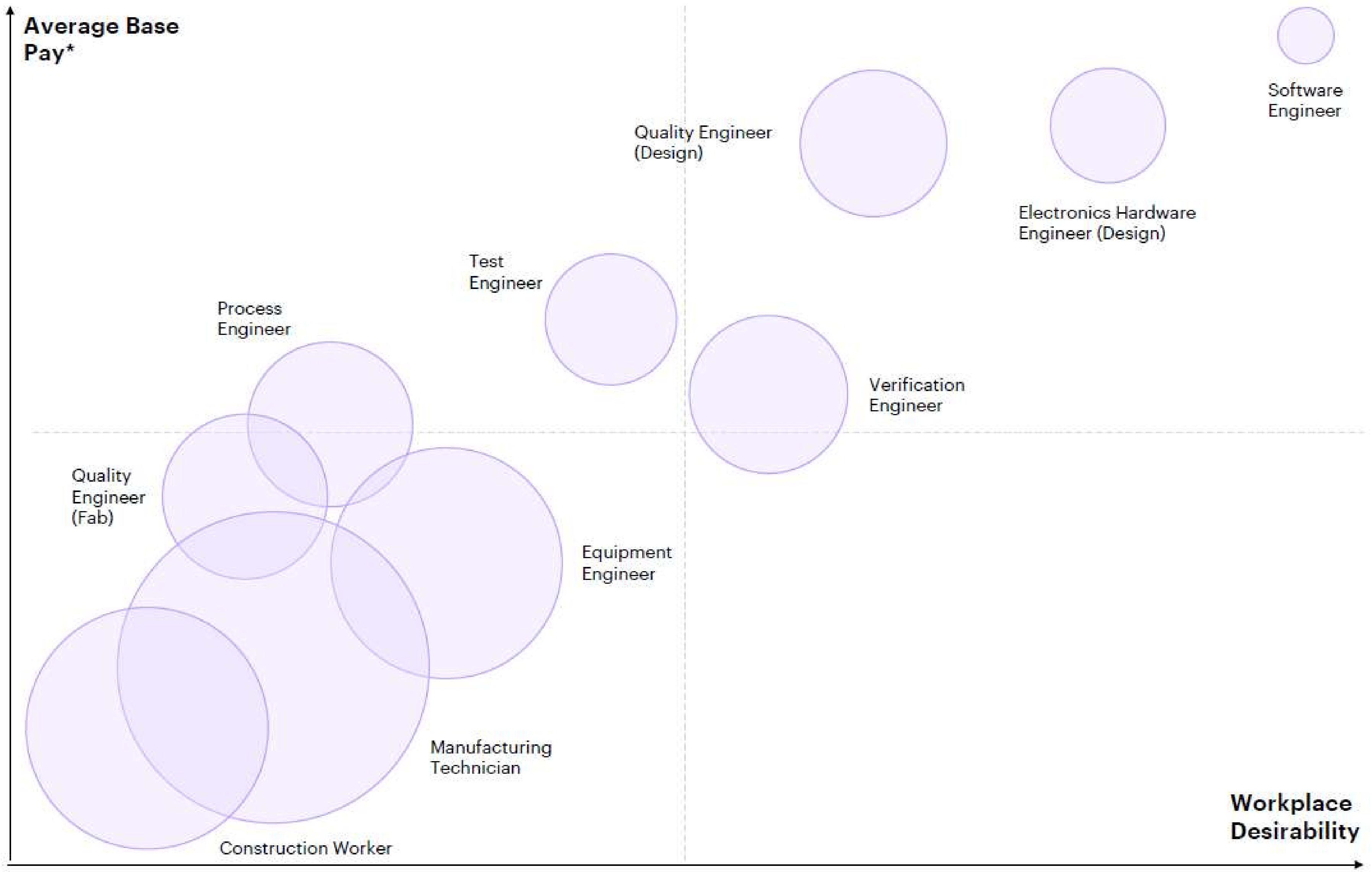
Semicon India Program



Semiconductor Workforce: Manufacturing Ecosystem



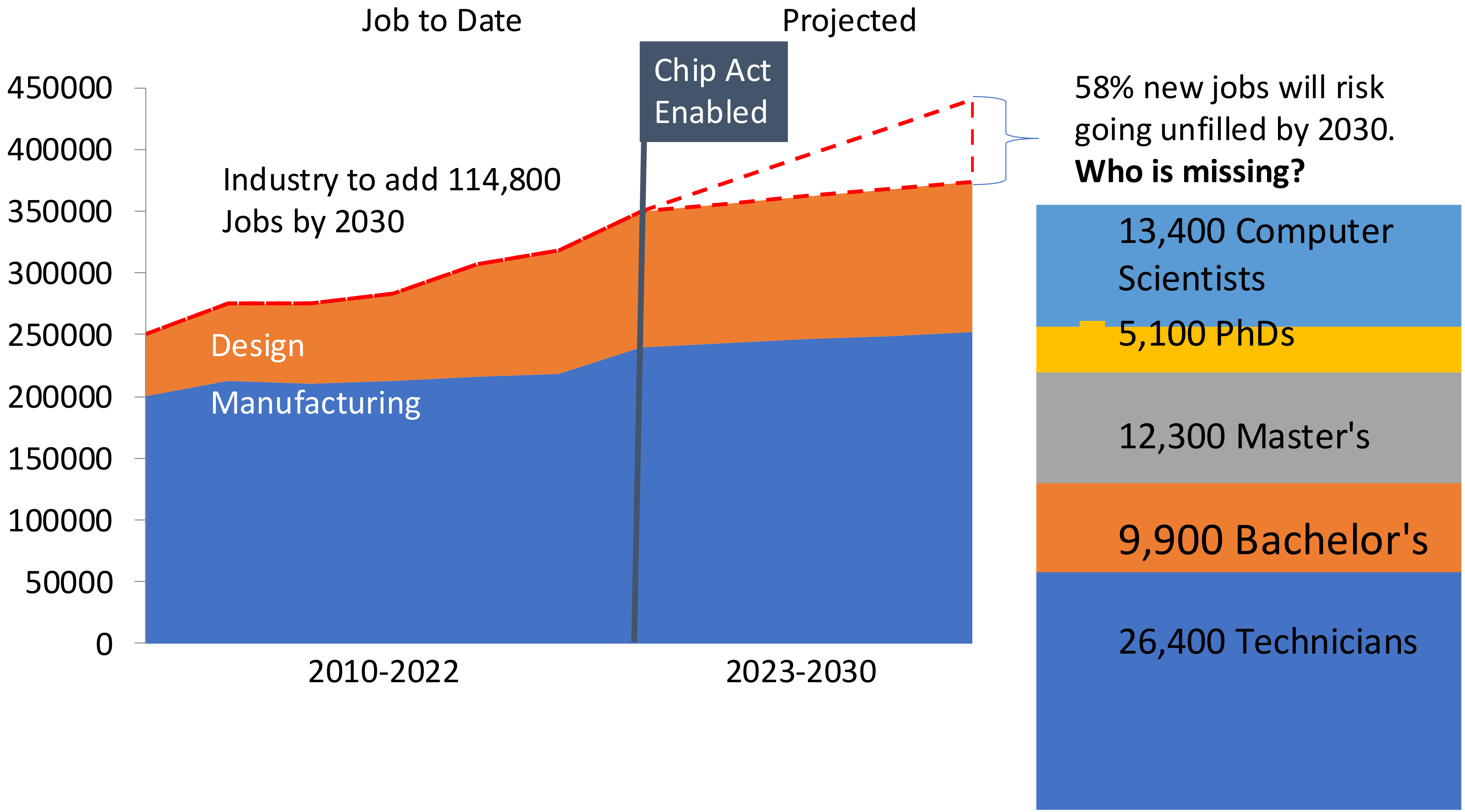
The relationship between desirability and demand
(Bubble size = Predicted talent shortage)



Source: Accenture Analysis

Semiconductor Workforce: Global Demand

Importance of Workforce development & Hands on Training



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Workforce Development Semiconductor Industry

- ✓ Prioritize talent as a strategic objective.
- ✓ Attract top engineering and scientific talent.
- ✓ Train a skilled workforce.
- ✓ Focus on technicians and community college programs.
- ✓ Utilize local, state programs.

Semiconductor Manufacturing: What's it All About?

Importance of Workforce development & Hands on Training

- Cleanroom environment
- High capital intensity
- Complex process technologies
- Ultra pure materials
- Advanced equipment
- High level of automation
- Tight process control
- Constant technical innovation

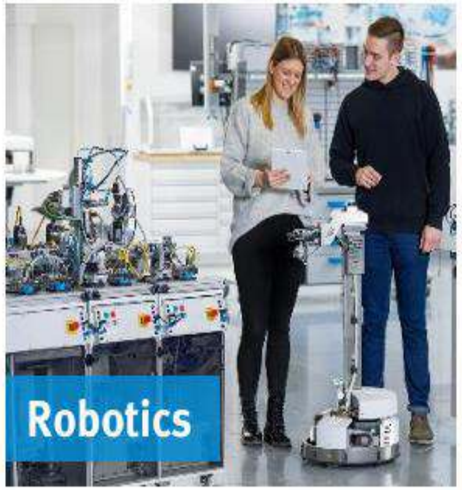
Demand Profile on Production floor

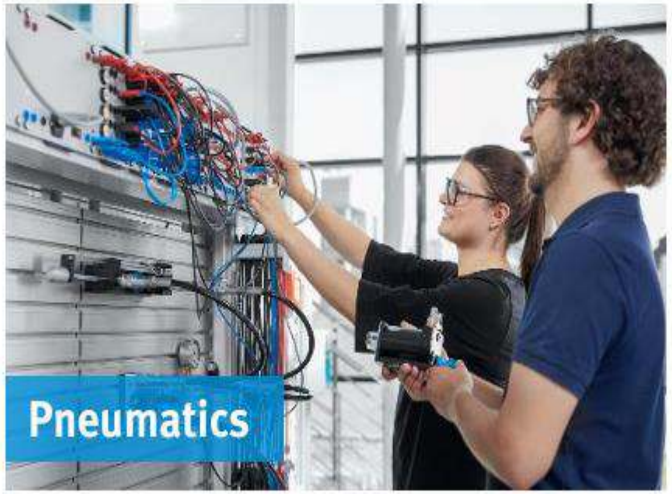
- Manufacturing/production technician
- Maintenance/equipment technician

Workforce: 2 Demand Profile on Production floor

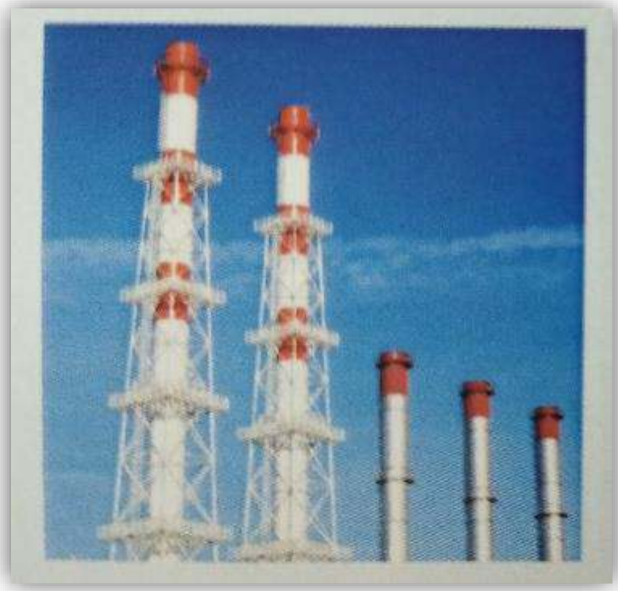
Both Type Need a Sound Skill set Spanning across following fields

\+ Soft skills





Semiconductor Workforce Skilling Areas: Utilities & Facilities



- DI Water Management; Bulk Gas Management;
- Analytical measurement of High Purity Gas and Chemical Distribution Network; Haz Chemical & Gas Handling and Management; PVAC/CDA

Environmental Monitoring
Fab Inner and Outer Environment; Safety Measures and Protocols

Waste Management (Effluent Treatment)

Clean Room: Filtration/HVAC/ CR Environment Control

GAP analysis for Semiconductor Workforce Development

Existing Gaps in India's Workforce

Workforce imbalance between design-side & manufacturing-side	Absence of Hands-on Training (OJTs)	Need to align Academic Output with Industry	Lack of Continuous Learning: Skilling, reskilling and upskilling	Lack of advanced R&D infrastructure

Potential Skill Development Opportunities

Semiconductor Curriculum

- **Industry-aligned** training curriculum
- **Train-the Trainer** - Deploy trained masters to Indian institutes
- Bridge courses for **Professionals Seeking to transition** into Semiconductor industry.

Provide Hand-on Training

- Establish **State of art-standard labs/ CoEs/ Semiconductor Research Centre** in Indian Technical institutes
- Bi-directional talent sharing and Industry internship **exchange program**

Build Advanced Research Infrastructure

Joint R&D centres /Center of Excellence:

- For **next-gen semiconductor** technologies
- Build **India focused products**

Skillset Requirement for Semiconductor Manufacturing

Current Landscape in India

1. India accounts for **20-25% of** the global **semiconductor design talent**

2. **Global Talent:** Large technical talent pool with **English language proficiency**

3. Most Indian talent focused on design & software, rather than **process technology and high-volume manufacturing**

Skillset Requirements

Fabrication & Packaging Process

- Process Engineer
- Yield Engineer
- Mask Engineer
- Technology Transfer Engineer
- Equipment Engineer
- Process Integration Engineer
- Quality Control Engineer
- IC Packaging Engineer
- IC Test Engineer
- Metrology & Inspection Engineer

Facilities & Construction

- Facilities Engineer (power, gas, water, effluent, hook-up, HVAC)
- Construction Supply Chain

Design Enablement

- TCAD Engineer
- PDK Engineer
- Design Enablement Engineer
- Physical Design Engineer (for customer support)

Smart Manufacturing

- Digital and Industry 4.0 skills
- Fab Automation Engineer
- Fab AI/ML Engineer
- Industrial Engineer
- Fab Q&R

Business Development

- Strategic Planning
- Foundry & Packaging Business Development
- Customer & Partner Engagement
- Market Intelligence & Analysis

Infra Readiness in the HEIs and Research Center across India

IIT Mandi: L0 and L1

SCL, Mohali: L3, L4, and L5

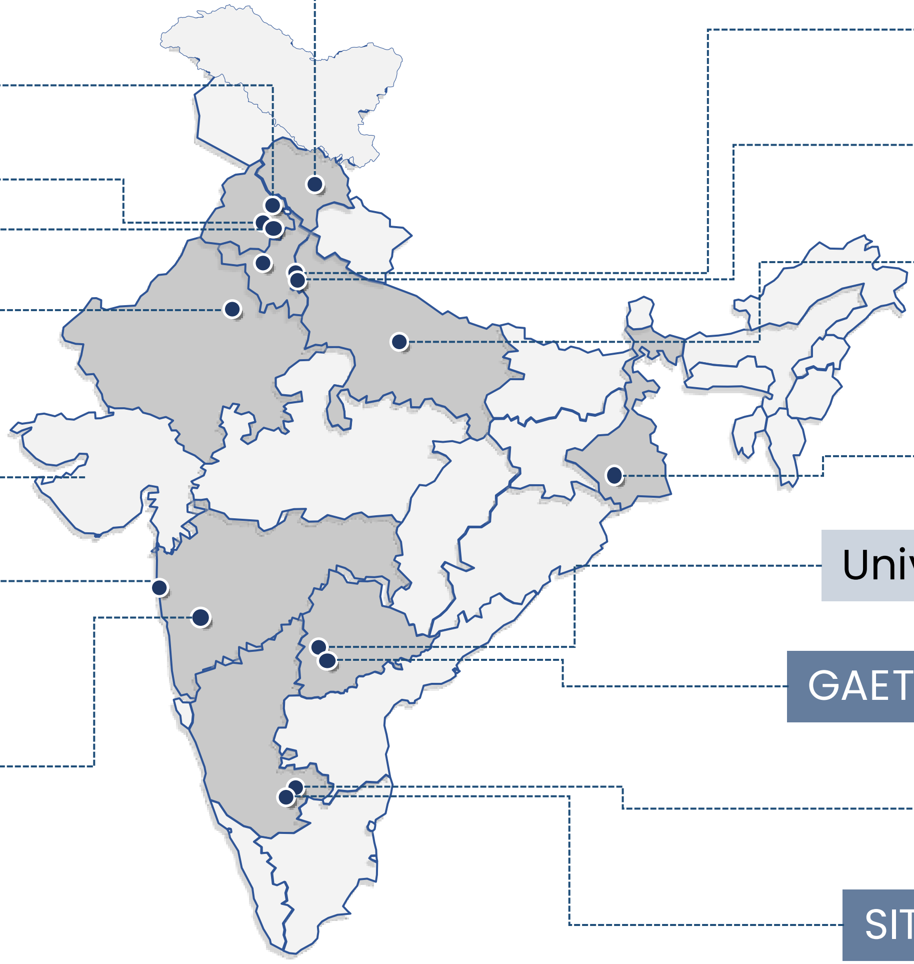
Punjab Engineering College: L0 and L1

IIT Ropar: L0

CEERI Pilani: L0 and L1

IIT Ghandhi Nagar L0, L1 and L2 & PDPU, Gati Shakti

IIIT Pune: L0 and L1



SSPL Delhi: L0 and L1

IIT Delhi: L0 and L1

IIT Kanpur: L0 and L1

IIT Kharagpur: L0 and L1

University of Hyderabad: L0 and L1

GAETEC Hyderabad: L0, L1, L2 and L3

IISc Bangalore : L0, L1, and L2

SITAR Bangalore : L0, L1, L2 and L3

L0	Basic Equipment for Micro Fabrication Facility
L1	Basic Equipment with Clean Room for Micro Fabrication Facility
L2	Prototyping Micro Fabrication Facility
L3	Fundamental Training on Micro Fabrication
L4	Advanced Training on Micro Fabrication
L5	Industry Ready Training on Micro Fabrication

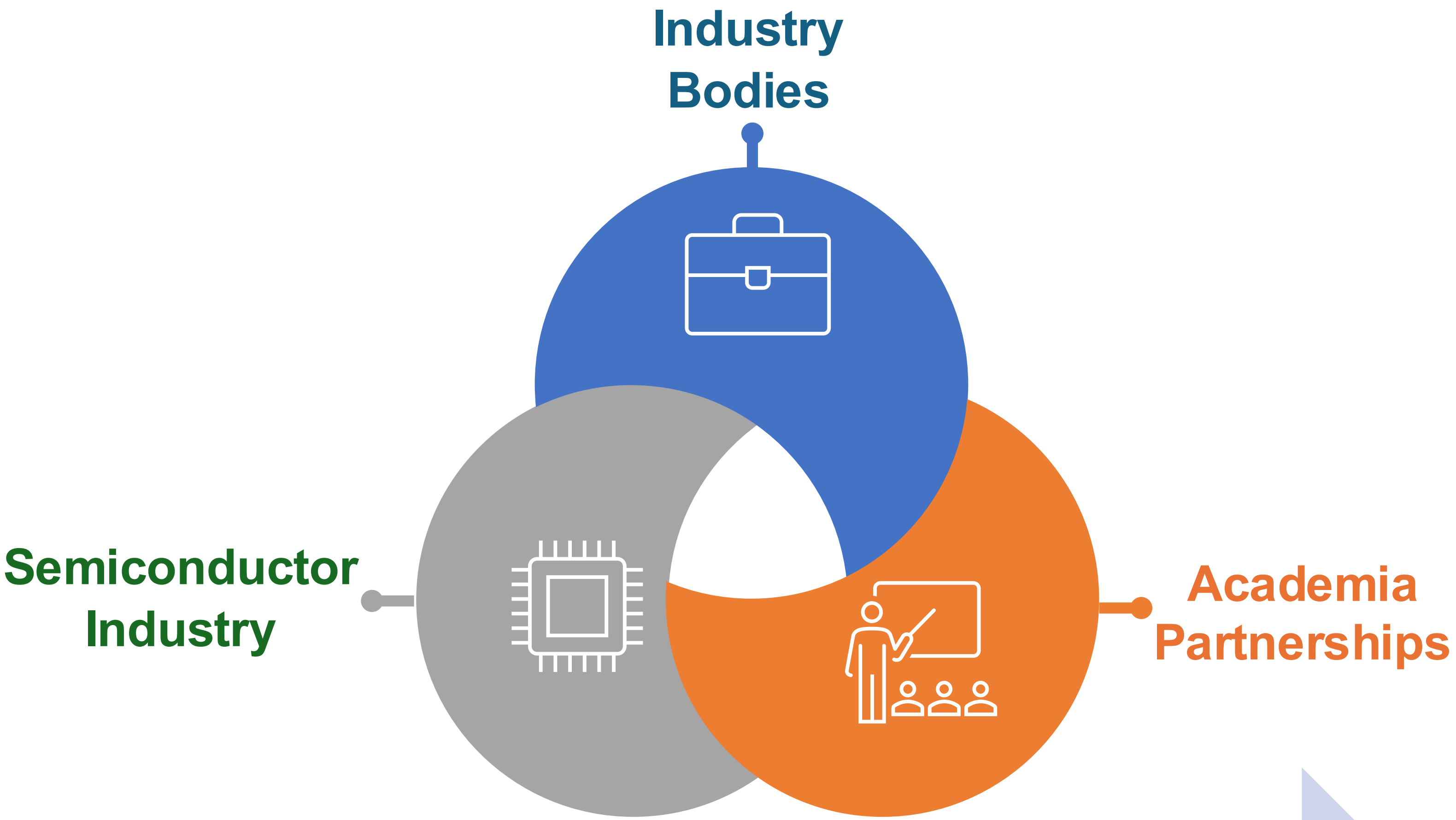
L5 Industry Ready — L0 Basic

Five actions to kickstart talent transformation

Prioritizing five actions will help semiconductor companies kickstart their talent transformations

1. **Lead with a new, compelling brand promotion**
2. **Nurture current talent by changing workplace culture**
3. **Expand the talent profiles**
4. **Leverage the ecosystem to plan for “known unknowns”**
5. **Reinvent the human capital function**

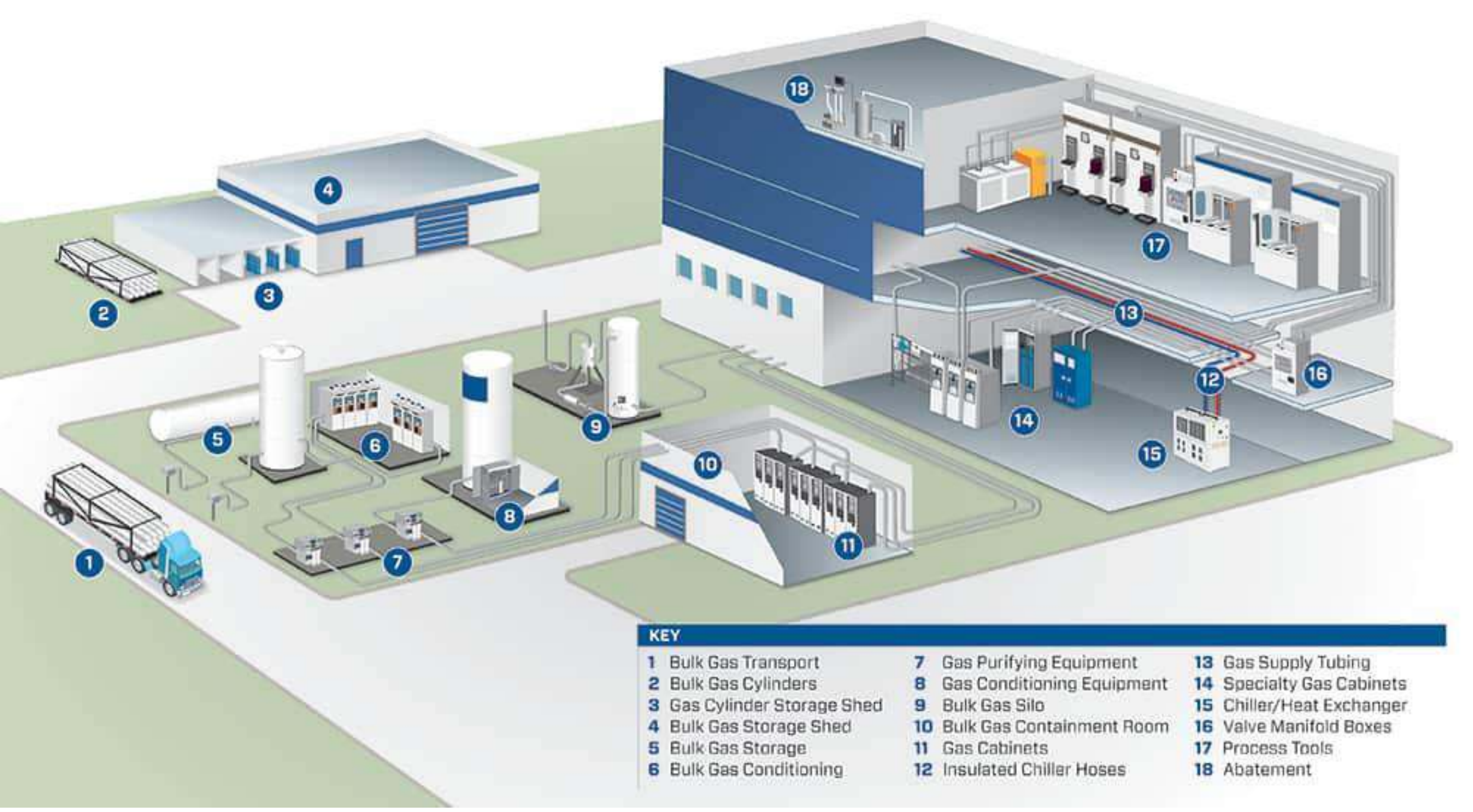
Approach for Workforce Development: Collaboration is Crucial



Collaboration drives a thriving and competitive ecosystem, fostering innovation, talent, and industry growth.

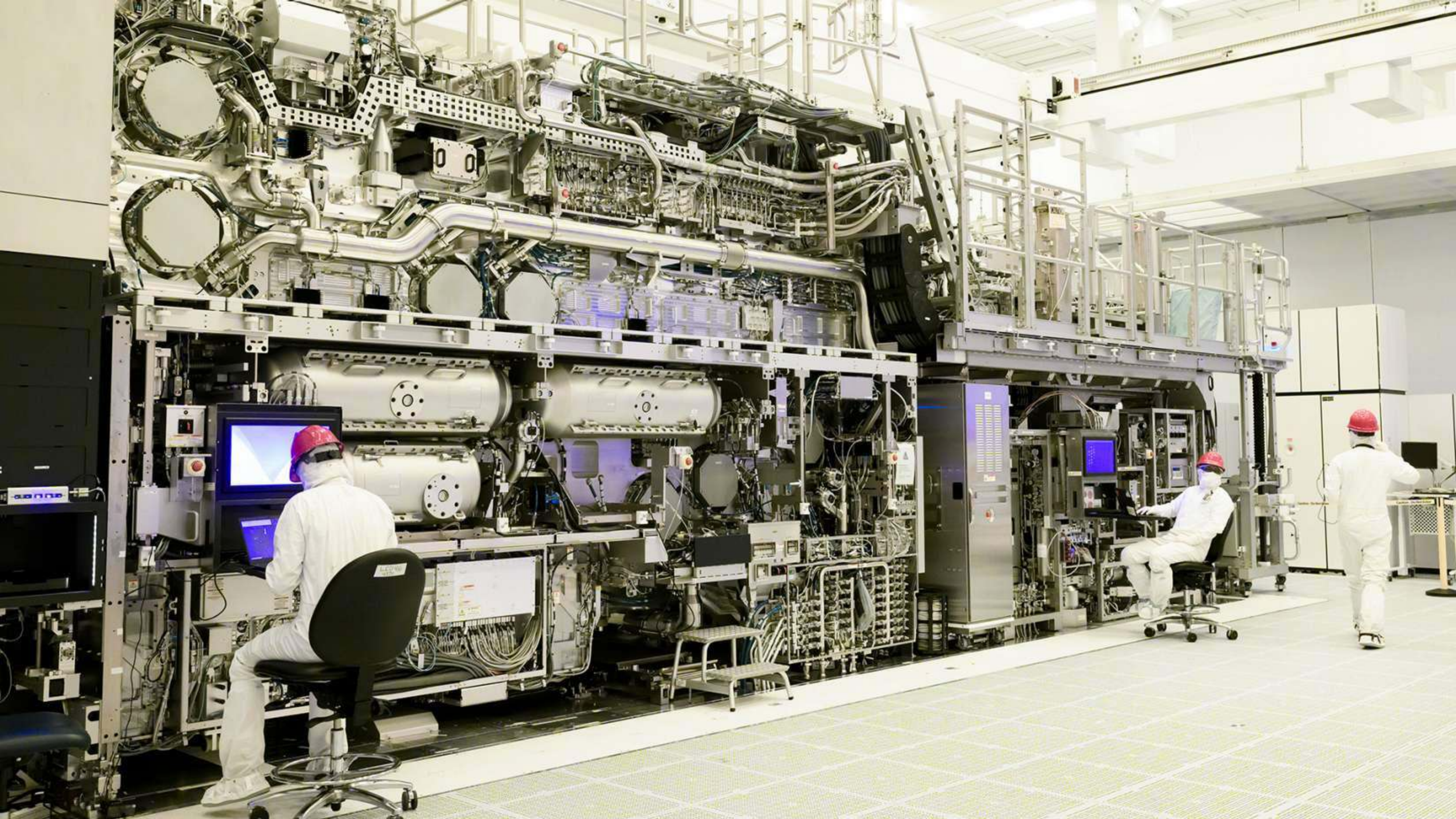
- Knowledge Exchange And Research
- Talent Development And Education
- Networking And Professional Development
- Innovation And Commercialization
- Policy & Regulation

जय हिन्द



KEY
1 Bulk Gas Transport
2 Bulk Gas Cylinders
3 Gas Cylinder Storage Shed
4 Bulk Gas Storage Shed
5 Bulk Gas Storage
6 Bulk Gas Conditioning
7 Gas Purifying Equipment
8 Gas Conditioning Equipment
9 Bulk Gas Silo
10 Bulk Gas Containment Room
11 Gas Cabinets
12 Insulated Chiller Hoses
13 Gas Supply Tubing
14 Specialty Gas Cabinets
15 Chiller/Heat Exchanger
16 Valve Manifold Boxes
17 Process Tools
18 Abatement


ASML



Workforce Skilling Areas

Talent Committee report-2022 (under AICTE)

Process Engineer

Education Domain: Electronics/chemical/ Electrical Engineering or materials science.

Skill Domain: Deep understanding of semiconductor manufacturing processes; Recipe Development; Problem-solving, data analysis, Defects analysis, DSA and Yield Learning,

Equipment Engineer

Education Domain: Electrical/Mechanical /Instrumentation/Electronics Engineering.

Skill Domain:

- Preventive Maintenance, Corrective Maintenance
- Troubleshoot, Equipment Health Checks
- Equipment Upgradation with OEM.

Device Engineer/Process Integration

Education Domain: Electronics/ semiconductor device physics

Skill Domain:

- Semiconductor design software Cadence, Synopsys, or Mentor Graphics.
- Strong understanding of semiconductor physics and fabrication processes.
- Experience with circuit simulation tools and methods.
- Problem-solving skills;
- Cross-functional collaboration

Quality Assurance

Education Domain : Industrial Engineering/Quality Management

Skill Domain

- Qualification and validation of Raw Materials
- Process Qualification
- Design experiments as per Process requirements,
- Implement statistical process control techniques, Understanding of Manufacturing Process,
- Product quality.

Workforce Skilling Areas (Talent Committee report-2022)

Safety and Hazard (ERT)

Education Domain: Fire and Safety/Chemical Engineering/ Environmental Science

Skill Domain:

- Fire and explosion prevention
- Safe manufacturing facilities
- Personal safety
- Compliance with applicate codes and standards
- Flash point
- Autoignition temperature
- Temperature limits of flammability
- Limits of flammability (lower and upper flammability limits) in air
- Limiting oxidant concentration (LOC)
- Conductivity
- Vapor pressure
- Boiling point & Melting point

Other Safety Training Measures

Education Domain : Process safety

Skill Domain :

- Modified flash point in a dry, H_2O-free atmosphere
- Limits of flammability at elevated temperatures and/or pressures
- Limits of flammability in alternate oxidizers such as O_2, N_2O, Cl_2, & F_2
- Limiting oxidant concentration (LOC) at elevated temp/pressure
- Limiting oxidant concentration with alternate oxidizers such as O_2, N_2O, Cl_2, and F_2
- Chemical compatibility testing in simulated process gases, including O_2, N_2O, NF_3, Cl_2, F_2 and NH_3
- Customized, large-scale testing

Workforce Skilling Areas (Talent Committee report-2022)

Quality Assurance Specialists

- Degree in Electrical Engineering, Quality Engineering, or a related field is a foundational requirement.
- Possessing knowledge of quality management systems, methodologies, and tools specific to IC Packaging.
- Understanding statistical analysis and quality control techniques
- Failure analysis.
- Quality standards and processes for ensuring product integrity and reliability.

Equipment Technicians

- Technical diploma in Electronics, Electrical Engineering Technology, or a related field.
- Demonstrating proficiency in operating and maintaining semiconductor testing, Assembly or wafer Fab equipment
- Possessing troubleshooting skills to diagnose and resolve equipment malfunctions
- knowledge of equipment calibration procedures and preventive maintenance is important for equipment reliability and accuracy.
- Understanding safety protocols and adhering to cleanroom practices

Workforce Skilling Areas (Talent Committee report-2022)

IC Packaging Engineer

- **Package semiconductor Technologies:** flip-chip, wire bonding, System-in-Package (SiP) and Fan-out Wafer Level Packaging
- **Advance Packaging:** 2.5D/3D/Heterogeneous Packaging
- Substrate and Materials for Packaging
- CAD software for package design.
- Thermal control, electrical performance, and reliability regarding IC packaging.

IC Test Engineer

- Degree in Electrical Engineering, Electronics, or a related field serves as a fundamental requirement.
- Test methodologies, test equipment handling, and test program development
- Possessing knowledge of semiconductor device physics and integrated circuit functionality.
- Exposing various testing techniques, such as functional testing, parametric testing, and burn-in testing
- Analytical and problem-solving skills specifically related to the Semiconductor Chip Packaging Industry

Workforce Skilling Areas

AICTE Approved Courses

Degree Courses

- **Electronics (VLSI Design and Technology)**
- **Introduction to IC Design & Technology**
- **Digital Systems Labs**
- **Semiconductor Device Fundamentals**
- **Analog Electronics**
- **Introduction to CMOS processing**
- **Fabrication and Characterization Lab**
- **Physics of Electrical Engineering Materials**

Diploma Courses

- Diploma in IC Manufacturing
- Introduction to VLSI Fabrication
- Semiconductor Fab Familiarization
- Electronic Devices and Circuits
- Clean Room Technologies
- Semiconductor Technology Equipment Maintenance

  Foundry like Safety Protocol for Foundry, Vacuum

  Technology Industrial Automation
- Semiconductor packaging and testing, electronics

  system assembly or product design